

Ayan Mondal

Master of Computer Applications (2025–2028)

National Institute of Technology Kurukshetra

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Summary

Motivated MCA student with hands-on experience in Generative AI, AI Agents, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, FastAPI, Spring Boot and full-stack application development. Experienced in designing scalable AI-powered systems, REST APIs, and containerized deployments using Docker.

Education

- **Master of Computer Applications**, National Institute of Technology Kurukshetra (2025–2028)
CGPA: 6.66/10
- Class XII: 79%
- Class X: 81%

Experience

AI Intern, VGI Skill Lab

(3 Months)

- Developed AI-powered applications using Python, FastAPI, LangChain, LangGraph, PostgreSQL and ChromaDB.
- Built enterprise-oriented Generative AI solutions utilizing Retrieval-Augmented Generation (RAG), AI Agents and semantic search.
- Designed RESTful APIs, integrated vector databases, and collaborated using Git-based development workflows.
- Containerized applications using Docker for streamlined deployment and portability.

Projects

SentinelDoc

- Developed an AI-powered document intelligence platform using Python, FastAPI, LangChain, PostgreSQL and ChromaDB.
- Implemented Retrieval-Augmented Generation (RAG) for context-aware document question answering over uploaded documents.
- Integrated vector embeddings and semantic search to improve retrieval accuracy.
- Dockerized the application and published container images to Docker Hub for simplified deployment.
- GitHub: <https://github.com/909ayanmondal-stack/SentinelDoc>

Software Engineering Multi-Agent Project

- Built a multi-agent AI system using LangGraph and Large Language Models (LLMs) to automate software engineering workflows.
- Designed specialized AI agents for planning, task execution and workflow orchestration.
- Implemented modular backend architecture with REST API integration for scalable agent communication.
- Leveraged prompt engineering and agent collaboration to improve task automation.
- GitHub: https://github.com/909ayanmondal-stack/Software_Eng._Multi_Agent_Project

Museum Management System (Backend)

- Developed a scalable backend for a Museum Management System using Spring Boot, Spring Data JPA and MySQL.
- Designed and implemented RESTful APIs for managing museums, exhibits, artifacts and users with complete CRUD functionality.
- Integrated relational database persistence using JPA/Hibernate and validated API functionality through comprehensive Postman testing.
- Followed a layered architecture with modular code organization to improve maintainability and scalability.
- Planned future enhancements including a React.js frontend, AI-powered RAG chatbot and Dockerized deployment.
- GitHub: https://github.com/909ayanmondal-stack/Museum_Backend

Crypto Hunter

- Developed a responsive cryptocurrency tracking web application using React.js.
- Integrated cryptocurrency market APIs to display real-time prices, market capitalization and historical market data.
- Implemented search, filtering and dynamic routing to provide an interactive user experience.
- Designed reusable React components with efficient state management and responsive UI.
- GitHub: https://github.com/909ayanmondal-stack/Crypto_Hunter

Technical Skills

Programming Languages: Python, Java, C, SQL

Generative AI: Large Language Models (LLMs), Generative AI, Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents, Multi-Agent Systems, Agentic AI, Embeddings, Semantic Search

Frameworks & Libraries: LangChain, LangGraph, FastAPI, Flask, Spring Boot, Spring Data JPA, Hibernate, React.js

Databases: PostgreSQL, MySQL, MongoDB

Developer Tools: Git, GitHub, Docker, Docker Hub, Postman, VS Code

Core Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), REST APIs, Software Engineering, System Design, Database Design